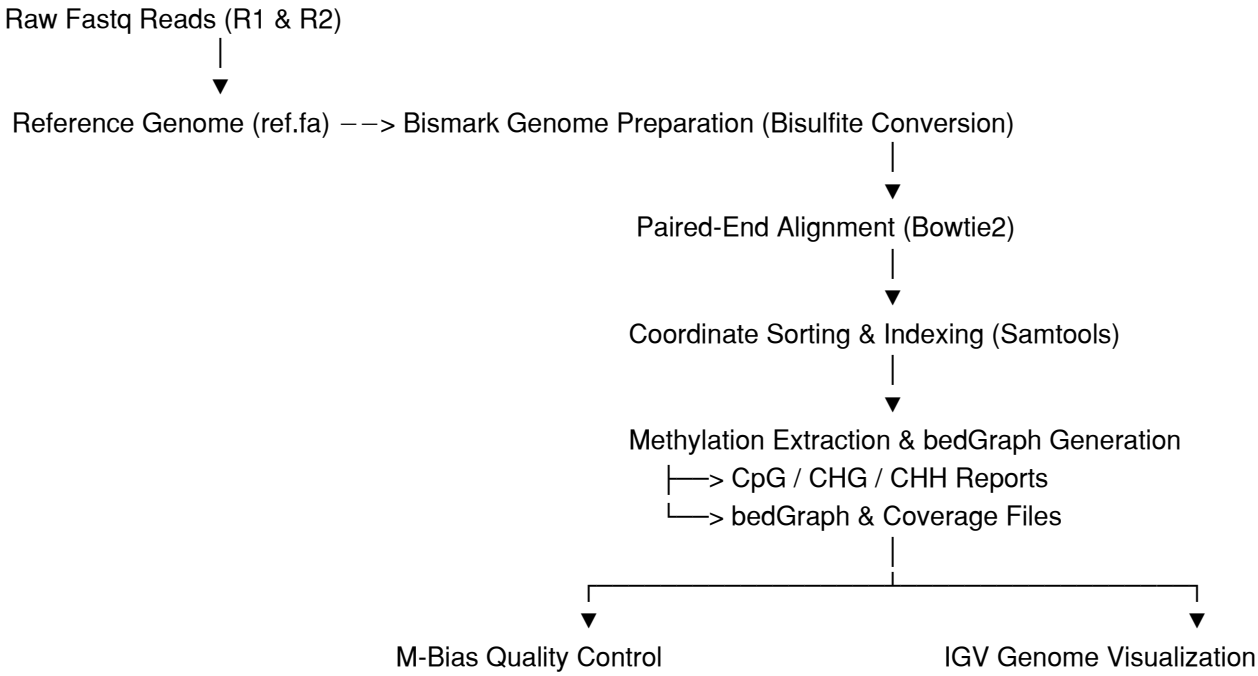
Name : Debarshi Das Adhikary
Date: 26/09/2023

Bisulfite Sequencing Analysis

1. Aim

To analyze Whole-Genome Bisulfite Sequencing (WGBS) data using the Bismark pipeline by preparing the bisulfite-converted reference genome, aligning paired-end sequencing reads, sorting and indexing BAM files, extracting cytosine methylation levels across CpG, CHG, and CHH contexts, generating bedGraph files for genomic visualization, and evaluating technical positional bias via M-bias and IGV profiling.

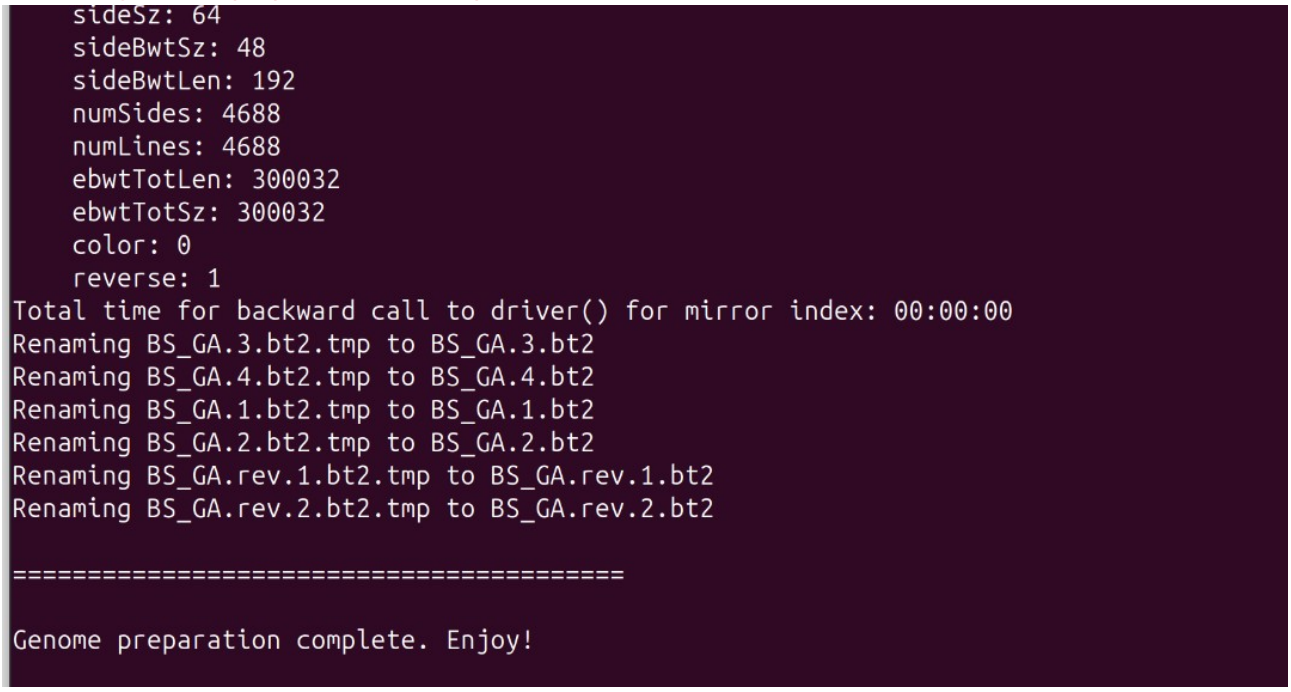
2. Analysis Workflow (Flowchart)



Step 1: Reference Genome Preparation
Bismark performs in silico bisulfite conversion of the reference genome and prepares indices to align bisulfite-treated reads, determine cytosine methylation status, and extract sequence contexts (CpG, CHG, CHH).

1. Create a directory named BS_seq_WGS and place the reference genome file (ref.fa) inside:

   Bash
   ```
   mkdir BS_seq_WGS
   cp ref.fa BS_seq_WGS/
   ```
2. Run the Bismark genome preparation tool:

   Bash
   ```
   bismark_genome_preparation BS_seq_WGS/
   ```
3. 
   ```
       sideSz: 64
       sideBwtSz: 48
       sideBwtLen: 192
       numSides: 4688
       numLines: 4688
       ebwtTotLen: 300032
       ebwtTotSz: 300032
       color: 0
       reverse: 1
   Total time for backward call to driver() for mirror index: 00:00:00
   Renaming BS_GA.3.bt2.tmp to BS_GA.3.bt2
   Renaming BS_GA.4.bt2.tmp to BS_GA.4.bt2
   Renaming BS_GA.1.bt2.tmp to BS_GA.1.bt2
   Renaming BS_GA.2.bt2.tmp to BS_GA.2.bt2
   Renaming BS_GA.rev.1.bt2.tmp to BS_GA.rev.1.bt2
   Renaming BS_GA.rev.2.bt2.tmp to BS_GA.rev.2.bt2

   ========================================

   Genome preparation complete. Enjoy!
   ```

Step 2: Bisulfite Alignment

Align single-end or separated paired reads (R1.fastq.gz and R2.fastq.gz) to the bisulfite-converted reference genome using Bowtie2:

Bash
```
bismark --bowtie2 BS_seq_WGS/ BS_seq_WGS/R1.fastq.gz
```

```
[WARNING] Failed to launch x86-64-v3 version, staying with default
46363 reads; of these:
  46363 (100.00%) were unpaired; of these:
    38146 (82.28%) aligned 0 times
    8191 (17.67%) aligned exactly 1 time
    26 (0.06%) aligned >1 times
17.72% overall alignment rate
46363 reads; of these:
  46363 (100.00%) were unpaired; of these:
    17598 (37.96%) aligned 0 times
    28659 (61.81%) aligned exactly 1 time
    106 (0.23%) aligned >1 times
62.04% overall alignment rate
Mapping summary for BS_seq_WGS/R1.fastq.gz:
sequences analysed:       46363
unique best alignments:   36912
no alignment found:       9408
ambiguous (unsuitable):   43
directional-rejected:     0
could-not-extract genomic:0
strand OT/OB/CTOT/CTOB:   8195/28717/0/0
(full report layout is Phase 6)
```

bismark --bowtie2 BS_seq_WGS/ BS_seq_WGS/R2.fastq.gz

```
[WARNING] Failed to launch x86-64-v3 version, staying with default
46363 reads; of these:
  46363 (100.00%) were unpaired; of these:
    46354 (99.98%) aligned 0 times
    9 (0.02%) aligned exactly 1 time
    0 (0.00%) aligned >1 times
0.02% overall alignment rate
46363 reads; of these:
  46363 (100.00%) were unpaired; of these:
    46352 (99.98%) aligned 0 times
    11 (0.02%) aligned exactly 1 time
    0 (0.00%) aligned >1 times
0.02% overall alignment rate
Mapping summary for BS_seq_WGS/R2.fastq.gz:
sequences analysed:       46363
unique best alignments:   20
no alignment found:       46343
ambiguous (unsuitable):   0
directional-rejected:     0
could-not-extract genomic:0
strand OT/OB/CTOT/CTOB:   11/9/0/0
(full report layout is Phase 6)
```

- **Output:** Generates mapping alignment BAM files (R1_bismark_bt2.bam, R2_bismark_bt2.bam) and alignment summary report text files.
- **Summary report for R1:**
- Final Alignment report
- ==========
- Sequences analysed in total: 46363
- Number of alignments with a unique best hit from the different alignments: 36912
- Mapping efficiency: 79.6%
- Sequences with no alignments under any condition: 9408

- Sequences did not map uniquely: 43
- Sequences which were discarded because genomic sequence could not be extracted: 0
-
- Number of sequences with unique best (first) alignment came from the bowtie output:
- CT/CT: 8195 ((converted) top strand)
- CT/GA: 28717 ((converted) bottom strand)
- GA/CT: 0 (complementary to (converted) top strand)
- GA/GA: 0 (complementary to (converted) bottom strand)
-
- Number of alignments to (merely theoretical) complementary strands being rejected in total: 0
-
- Final Cytosine Methylation Report
- ==========
- Total number of C's analysed: 875967
-
- Total methylated C's in CpG context: 29030
- Total methylated C's in CHG context: 851
- Total methylated C's in CHH context: 2211
- Total methylated C's in Unknown context: 3
-
- Total unmethylated C's in CpG context: 17594
- Total unmethylated C's in CHG context: 222073
- Total unmethylated C's in CHH context: 604208
- Total unmethylated C's in Unknown context: 532
-
- C methylated in CpG context: 62.3%
- C methylated in CHG context: 0.4%
- C methylated in CHH context: 0.4%
- C methylated in Unknown context (CN or CHN): 0.6%

Summary report for R2:

Final Alignment report
==========
Sequences analysed in total: 46363
Number of alignments with a unique best hit from the different alignments: 20
Mapping efficiency: 0.0%
Sequences with no alignments under any condition: 46343
Sequences did not map uniquely: 0
Sequences which were discarded because genomic sequence could not be extracted: 0

Number of sequences with unique best (first) alignment came from the bowtie output:
CT/CT: 11 ((converted) top strand)
CT/GA: 9 ((converted) bottom strand)
GA/CT: 0 (complementary to (converted) top strand)

GA/GA: 0 (complementary to (converted) bottom strand)

Number of alignments to (merely theoretical) complementary strands being rejected in total: 0

Final Cytosine Methylation Report
==========
Total number of C's analysed: 485

Total methylated C's in CpG context: 10
Total methylated C's in CHG context: 8
Total methylated C's in CHH context: 282
Total methylated C's in Unknown context: 1

Total unmethylated C's in CpG context: 2
Total unmethylated C's in CHG context: 40
Total unmethylated C's in CHH context: 143
Total unmethylated C's in Unknown context: 0

C methylated in CpG context: 83.3%
C methylated in CHG context: 16.7%
C methylated in CHH context: 66.4%
C methylated in Unknown context (CN or CHN): 100.0%

Step 3: Sort the BAM Files

Sort the resulting alignment files by genomic coordinates using Samtools:

Bash

```
samtools sort R1_bismark_bt2.bam R1_bismark_bt2_sorted
samtools sort R2_bismark_bt2.bam R2_bismark_bt2_sorted
```

Step 4: Index the Sorted BAM Files

Generate index files (.bai) required for fast random access and visualization in IGV:

```
samtools index R1_bismark_bt2_sorted.bam
samtools index R2_bismark_bt2_sorted.bam
```

Step 5: Methylation Extraction and bedGraph Generation

Extract context-dependent cytosine methylation calls and generate genome-wide bedGraph coverage files:

Bash

```
bismark_methylation_extractor --bedGraph --counts R1_bismark_bt2_sorted.bam
```

Total number of C's analysed: 875967

Total methylated C's in CpG context: 29030
Total methylated C's in CHG context: 851
Total methylated C's in CHH context: 2211

Total C to T conversions in CpG context: 17594
Total C to T conversions in CHG context: 222073
Total C to T conversions in CHH context: 604208

C methylated in CpG context: 62.3%
C methylated in CHG context: 0.4%
C methylated in CHH context: 0.4%

bismark_methylation_extractor --bedGraph --counts R2_bismark_bt2_sorted.bam

Final Cytosine Methylation Report
==========
Total number of C's analysed: 485

Total methylated C's in CpG context: 10
Total methylated C's in CHG context: 8
Total methylated C's in CHH context: 282

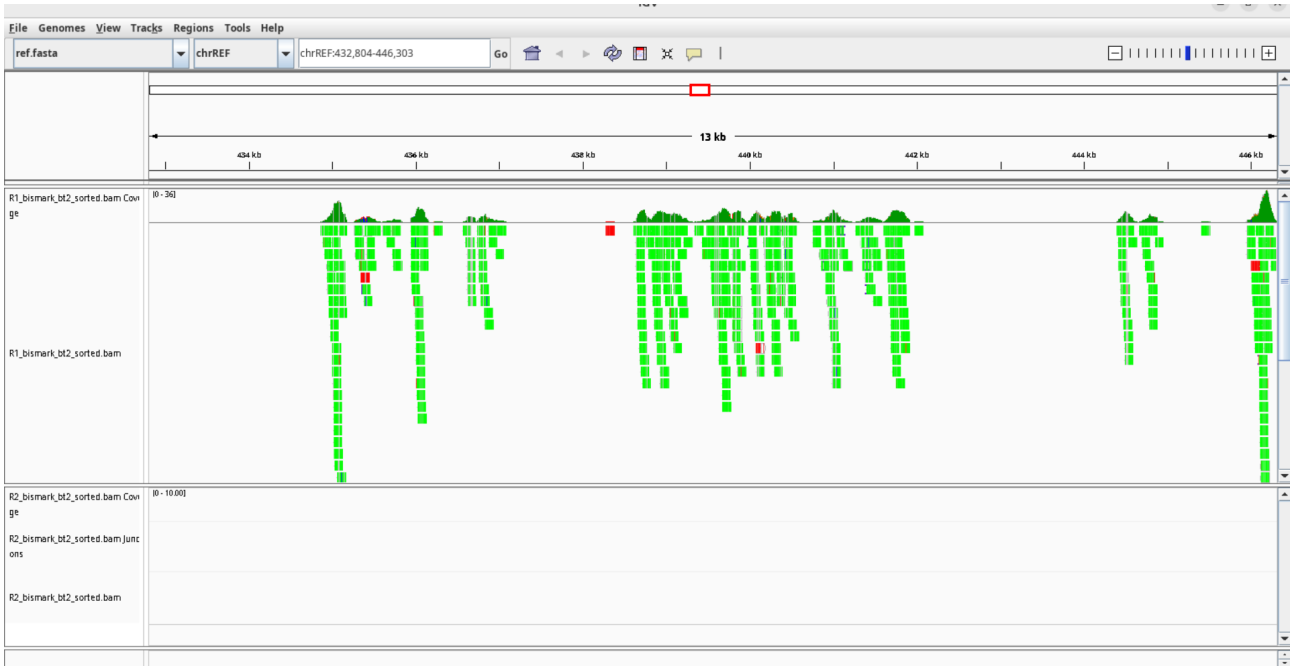
Total C to T conversions in CpG context: 2
Total C to T conversions in CHG context: 40
Total C to T conversions in CHH context: 143

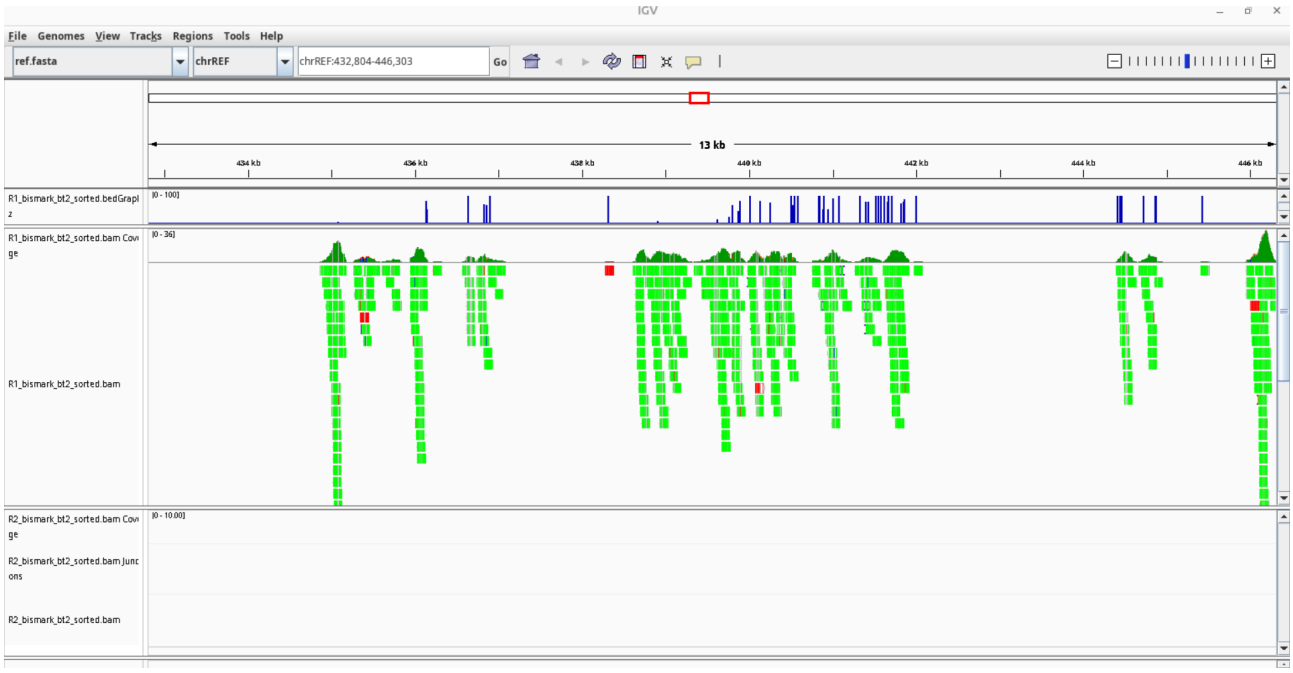
C methylated in CpG context: 83.3%
C methylated in CHG context: 16.7%
C methylated in CHH context: 66.4%

- **Output:** Generates bedGraph files, coverage files, and M-bias report files (.M-bias.txt).

3. Visualisation in IGV (Integrative Genomics Viewer)

Load the reference genome (ref.fa), the sorted BAM files (.bam), and the generated bedGraph files into IGV to evaluate coverage and methylation patterns.



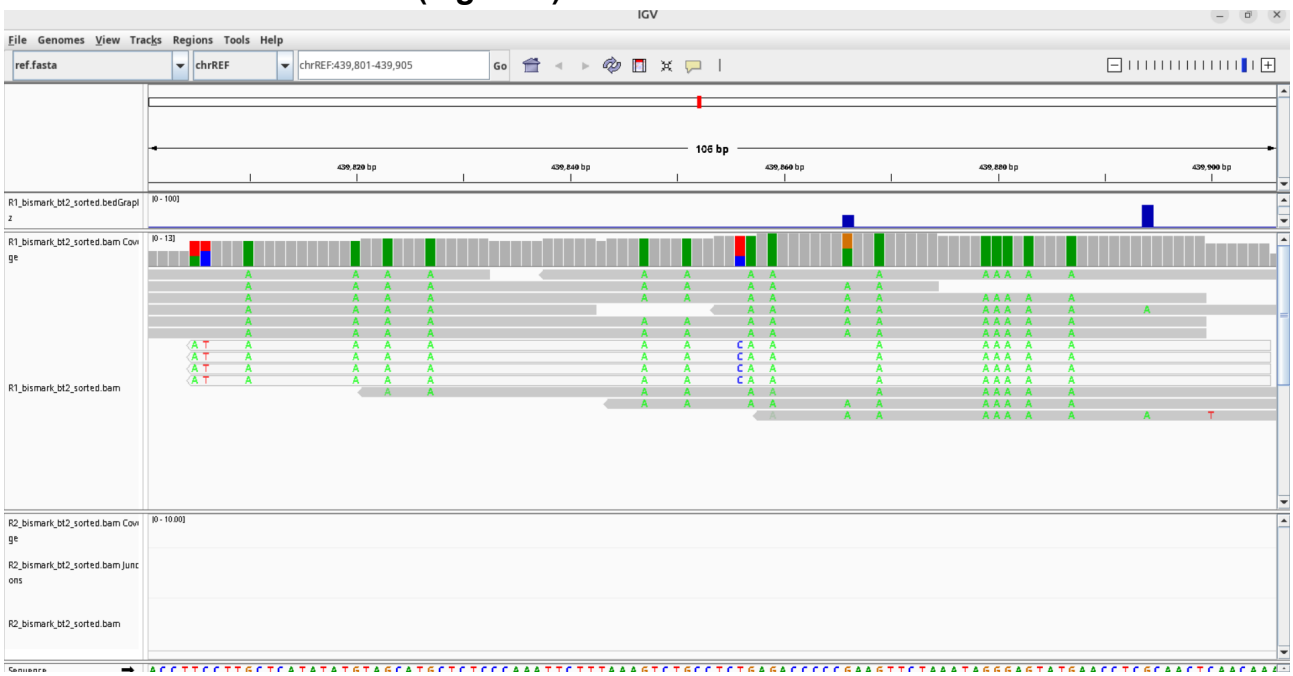


Broad Genomic Region Interpretation (Figure 1)

- **Blue Vertical Spikes (bedGraph Track):** Represent genomic positions where Bismark identified cytosine methylation. The height of each spike corresponds to the percentage of methylation at that site. Taller peaks indicate higher methylation percentages, while flat or empty regions denote zero signal or lack of reported methylation.

- **Green Coverage Peaks (BAM Track):** Represent sequencing read pileups. Increased vertical height reflects higher read depth and coverage across that genomic locus.

Base-Pair Resolution View (Figure 2)



- **Local Read Alignment:** The zoomed-in 106-bp interval shows individual R1 reads aligned to the reference sequence. Most bases match the reference genome, with a small number of mismatches (e.g., C-to-T transitions) visible.

- **Bisulfite Conversion Context:** C-to-T differences typically stem from unmethylated cytosines converted to uracil (read as thymine) during bisulfite treatment. However, individual base mismatches require Bismark context calls and strand orientation before being classified as methylated or unmethylated.

- **R1 vs. R2 Discrepancy:** The R1 track displays clear local coverage across this interval, whereas the R2 track exhibits no reads, showing lack of coverage for R2 at this specific locus rather than a paired-end alignment view.

4. Position-Wise Percentage Methylation (M-Bias Analysis)

M-bias analysis assesses whether methylation call percentages systematically vary according to the base position along the sequencing read.

CpG M-Bias Summary (First 17 Read Positions)

From inspection of `head -20 R1_bismark_bt2_sorted.M-bias.txt`:

Read Position	Count Methylated	Count Unmethylated	% Methylation	Total Coverage
1	314	172	64.18%	482
2	239	85	73.70%	327
3	233	127	64.72%	360
4	269	156	63.29%	425
5	241	157	60.55%	398
6	325	165	66.33%	490
7	272	128	68.00%	400
8	254	178	58.80%	432
9	284	164	63.39%	448
10	308	205	60.04%	513
11	289	168	63.24%	457
12	293	156	65.26%	449

Read Position	Count Methylated	Count Unmethylated	% Methylation	Total Coverage
13	282	168	62.67%	450
14	269	142	65.45%	411
15	251	159	61.22%	410
16	273	162	62.76%	435
17	326	157	67.49%	483

Interpretation

- Methylation levels across the first 17 positions range from **58.80%** (position 8) to **73.70%** (position 2).
- There is no obvious systematic upward or downward trend across these initial positions, suggesting that severe technical bias is not immediately apparent here.
- The full length of the read—particularly the extreme 5′ and 3′ ends—must still be examined to decide whether end-trimming of biased cycles is necessary before final methylation quantification.

5. Conclusion

- R1 and R2 sequencing reads were aligned independently to the in silico converted genome, followed by sorting and indexing with Samtools.
- Bismark extracted cytosine-level methylation calls and produced bedGraph tracks, allowing visual confirmation of read depth and methylation density across regions in IGV.
- CpG sites showed substantial methylation levels compared to non-CpG contexts (CHG, CHH).
- M-bias evaluation verified read-position consistency across early sequencing cycles, completing the quality control and mapping pipeline.